Q2. [16 pts] Performing PCA by Hand

Let's do principal components analysis (PCA)! Consider this sample of six points $X_i \in \mathbb{R}^2$.

$$\left\{\begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}2\\1\end{bmatrix}, \begin{bmatrix}2\\2\end{bmatrix}\right\}.$$

(a) [3 pts] Compute the mean of the sample points and write the centered design matrix $\dot{X}$.

The sample mean is

$$\mu = \frac{1}{6}\sum_{i=1}^{6} X_i = \begin{bmatrix}1\\1\end{bmatrix}.$$

By subtracting the mean from each sample, we form the centered design matrix

$$\dot{X} = \begin{bmatrix}-1 & -1\\ -1 & 0\\ 0 & -1\\ 0 & 1\\ 1 & 0\\ 1 & 1\end{bmatrix}.$$

(b) [6 pts] Find all the principal components of this sample. Write them as **unit** vectors.

The principal components of our dataset are the eigenvectors of the matrix

$$\dot{X}^\top \dot{X} = \begin{bmatrix}4 & 2\\ 2 & 4\end{bmatrix}.$$

The characteristic polynomial of this symmetric matrix is

$$\det(sI - X^\top X) = \det\begin{bmatrix}s-4 & -2\\ -2 & s-4\end{bmatrix} = (s-4)(s-4) - (-2)(-2)$$
$$= s^2 - 8s + 12 = (s-6)(s-2).$$

Hence the eigenvalues of $\dot{X}^\top \dot{X}$ are $\lambda_1 = 2$ and $\lambda_2 = 6$. With these eigenvalues, we can compute the eigenvectors of this matrix as follows. (Or you could just guess them and verify them.)

$$\begin{bmatrix}\lambda_1 - 4 & -2\\ -2 & \lambda_1 - 4\end{bmatrix} v_1 = \begin{bmatrix}-2 & -2\\ -2 & -2\end{bmatrix} v_1 = \begin{bmatrix}0\\0\end{bmatrix} \Rightarrow v_1 = \begin{bmatrix}1/\sqrt{2}\\ -1/\sqrt{2}\end{bmatrix}$$

$$\begin{bmatrix}\lambda_2 - 4 & -2\\ -2 & \lambda_2 - 4\end{bmatrix} v_2 = \begin{bmatrix}2 & -2\\ -2 & 2\end{bmatrix} v_2 = \begin{bmatrix}0\\0\end{bmatrix} \Rightarrow v_2 = \begin{bmatrix}1/\sqrt{2}\\ 1/\sqrt{2}\end{bmatrix}$$

(c) [4 pts]

- Which of those two principal components would be preferred if you use only one? [1 pt]
- What information does the PCA algorithm use to decide that one principal components is better than another? [1 pt]
- From an optimization point of view, why do we prefer that one? [2 pts]

We choose $v_2 = [1/\sqrt{2} \quad 1/\sqrt{2}]^\top$ first.

PCA picks the principal component with the largest eigenvalue.

We prefer it because it maximizes the variance of the sample points after they are projected onto a line parallel to v_2. (Alternatively, because it minimizes the sum of squares of distances from each sample point to its projection on that line.)

(d) [3 pts] Compute the **vector** projection of each of the **original** sample points (**not** the centered sample points) onto your **preferred principal component**. By "vector projection" we mean that the projected points are still in $\mathbb{R}^2$. (Don't just give us the principal coordinate; give us the projected point.)

The vector projection of X_i onto a direction w is

$$\tilde{X}_i = \frac{X_i^\top w}{\|w\|_2^2} w.$$

If w is a unit vector, we can just write $\tilde{X}_i = (X_i^\top w)w$.

For our sample and our chosen principal component $w = v_2$, the projections are

$$\tilde{X}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \tilde{X}_2 = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}, \tilde{X}_3 = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}, \tilde{X}_4 = \begin{bmatrix} 3/2 \\ 3/2 \end{bmatrix}, \tilde{X}_5 = \begin{bmatrix} 3/2 \\ 3/2 \end{bmatrix}, \tilde{X}_6 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Q4. [15 pts] Backpropagation on an Arithmetic Expression

Consider an arithmetic network with the inputs a, b, and c, which computes the following sequence of operations, where $s(\gamma) = \dfrac{1}{1+e^{-\gamma}}$ is the logistic (sigmoid) function and $r(\gamma) = \max\{0, \gamma\}$ is the hinge function used by ReLUs.

$$d = ab \quad e = s(d) \quad f = r(a) \quad g = 3a \quad h = 2e + f + g \quad i = ch \quad j = f + i^2$$

We want to find the partial derivatives of j with respect to every other variable a through i, in backpropagation style. This means that for each variable z, we want you to write $\partial j/\partial z$ in two forms: (1) in terms of derivatives involving each variable that *directly* uses the value of z, and (2) in terms of the inputs and intermediate values $a \ldots i$, as simply as possible but with no derivative symbols. For example, we write

$$\frac{\partial j}{\partial i} = \frac{\mathrm{d}j}{\mathrm{d}i} = 2i \quad \text{(no chain rule needed for this one only)}$$
$$\frac{\partial j}{\partial h} = \frac{\partial j}{\partial i}\frac{\partial i}{\partial h} = 2ic \quad \text{(chain rule, then backprop the derivative expressions)}$$

(a) [15 pts] Now, please write expressions for $\partial j/\partial g$, $\partial j/\partial f$, $\partial j/\partial e$, $\partial j/\partial d$, $\partial j/\partial c$, $\partial j/\partial b$, and $\partial j/\partial a$ as we have written $\partial j/\partial h$ above. If they are needed, express the derivative $s'(\gamma)$ in terms of $s(\gamma)$ and express the derivative $r'(\gamma)$ as the indicator function $\mathbf{1}(\gamma \geq 0)$. (*Hint: f is used in two places and a is used in three, so they will need a multivariate chain rule. It might help you to draw the network as a directed graph, but it's not required.*)

$$\frac{\partial j}{\partial g} = \frac{\partial j}{\partial h}\frac{\partial h}{\partial g} = 2ic$$
$$\frac{\partial j}{\partial f} = \frac{\partial j}{\partial h}\frac{\partial h}{\partial f} + \frac{\mathrm{d}j}{\mathrm{d}f} = 2ic + 1$$
$$\frac{\partial j}{\partial e} = \frac{\partial j}{\partial h}\frac{\partial h}{\partial e} = 4ic$$
$$\frac{\partial j}{\partial d} = \frac{\partial j}{\partial e}\frac{\partial e}{\partial d} = 4ic\, s(d)(1 - s(d))$$
$$\frac{\partial j}{\partial c} = \frac{\partial j}{\partial i}\frac{\partial i}{\partial c} = 2ih$$
$$\frac{\partial j}{\partial b} = \frac{\partial j}{\partial d}\frac{\partial d}{\partial b} = 4ica\, s(d)(1 - s(d))$$
$$\frac{\partial j}{\partial a} = \frac{\partial j}{\partial d}\frac{\partial d}{\partial a} + \frac{\partial j}{\partial f}\frac{\partial f}{\partial a} + \frac{\partial j}{\partial g}\frac{\partial g}{\partial a} = 4icb\, s(d)(1 - s(d)) + (2ic + 1) \cdot \mathbf{1}(a \geq 0) + 6ic$$

Q6. [20 pts] Quadratic Regression

You are given a design matrix $X \in \mathbb{R}^{n \times d}$, storing n training points with d features, and a vector $y \in \mathbb{R}^n$ of continuously-valued labels. As usual, let X_i be training point i, expressed as a column vector such that row i of X is $X_i^\top$. We want to perform least-squares regression with the quadratic regression function $h(z) = z^\top W z$ (no linear nor constant terms), where $W \in \mathbb{R}^{d \times d}$ is symmetric. The objective is to find the symmetric matrix W that minimizes the least-squares cost function

$$J(W) = \sum_{i=1}^{n} \left(X_i^\top W X_i - y_i \right)^2 .$$

(*Note: If you get stuck on one part, try the other parts; each part is independent of solving the others.*)

(a) [6 pts] Find $\nabla_W J(W)$. (*Hint: It should be a $d \times d$ matrix like W. If you don't remember how to take a derivative with respect to a matrix, try writing $X_i^\top W X_i$ as a summation, but your final answer should have just one summation, over i.*)

$$\begin{aligned}
\nabla_W J(W) &= \sum_{i=1}^{n} \nabla_W (X_i^\top W X_i - y_i)^2 \\
&= \sum_{i=1}^{n} 2(X_i^\top W X_i - y_i) \cdot \nabla_W (X_i^\top W X_i - y_i) \\
&= 2 \sum_{i=1}^{n} (X_i^\top W X_i - y_i) X_i X_i^\top .
\end{aligned}$$

(b) [4 pts] Suppose that X has rank n. (That is, the training points are linearly independent and $n \leq d$.) Let I_i be the **row vector** that is row i of the $n \times n$ identity matrix; hence, its entries are zero except a 1 at position i. Observe that we can write training point i as the row vector $X_i^\top = I_i X$. Let X^+ be the Moore–Penrose pseudoinverse of X. As a prelude to part (c), show that $X_i^\top X^+ = I_i$. (*Hint: What do you know about pseudoinverses?*)

By postmultiplying both sides of $X_i^\top = I_i X$ by X^+, we have $X_i^\top X^+ = I_i X X^+ = I_i$. (As X is an $n \times d$ matrix with rank n, $XX^+ = I$.)

(c) [5 pts] Again suppose that X has rank n. Write the labels as a diagonal $n \times n$ matrix

$$Y = \operatorname{diag}(y_i) = \begin{bmatrix} y_1 & 0 & \dots & 0 \\ 0 & y_2 & & 0 \\ \vdots & & \ddots & \\ 0 & \dots & & y_n \end{bmatrix}.$$

Using (b), show that the cost function $J(W)$ is minimized by $W^* = X^+ Y X^{+\top}$, where $X^{+\top}$ denotes the transpose of X^+.

For each $i \in [1, n]$,

$$X_i^\top W^* X_i = X_i^\top X^+ Y X^{+\top} X_i = I_i Y I_i^\top = y_i.$$

Therefore, $J(W^*) = \sum_{i=1}^{n} (X_i^\top W^* X_i - y_i)^2 = 0$. As $J(W)$ is nonnegative for all W, no matrix has a lower cost than W^*, so W^* is optimal.

Note: we will also accept answers that show that $\nabla_W J(W)$ is zero at $W = W^*$ (instead of showing that $J(W^*) = 0$).

(d) [5 pts] Suppose the ground truth for the labels is $g(z) = z^\top A z$, where $A \in \mathbb{R}^{d \times d}$ is a fixed, symmetric matrix. Each training label is drawn from this ground truth with random Gaussian noise; that is, $y_i = g(X_i) + \epsilon_i$ where $\epsilon_i \sim \mathcal{N}(0, 1)$. Assume that the regression function we learn is $h(z) = z^\top W^* z$ where W^* is given in part (c). Show that the **bias** of this regression method at a test point z is

$$\text{bias}(z) = |z^\top (X^+ \operatorname{diag}(X_i^\top A X_i) X^{+\top} - A) z|,$$

where $\text{diag}(X_i^\top A X_i)$ is a diagonal $n \times n$ matrix whose diagonal values are $X_i^\top A X_i$.

$$\text{bias}(z) = |E[h(z)] - g(z)| = |E[z^\top X^+ Y X^{+\top} z] - z^\top A z| = |z^\top (X^+ E[Y] X^{+\top} - A) z|.$$

$E[Y]$ is a diagonal matrix whose diagonal values are $E[y_i] = E[g(X_i) + \epsilon_i] = g(X_i) = X_i^\top A X_i$, so

$$\text{bias}(z) = |z^\top (X^+ \text{diag}(X_i^\top A X_i) X^{+\top} - A) z|.$$